Friday Lecture

We cover §3.2
See reading 3 for description

Recap

We can compare cardinalities using properties of function.

$|A| = |B|$ iff it is possible to find a bijection from A to B

$|A| \leq |B|$ iff it is possible to find an injection from A to B

This is justified by the fact
$\leq$ and $=$ behave as we expect

i.e
If $|A| \leq |B|$ and $|B| \leq |C|$ then $|A| \leq |C|$

Why If $|A| \leq |B|$, then there is an injection $f : A \to B$

If $|B| \leq |C|$, then there is an injection $g : B \to C$

Claim

$g \circ f$ is an injection

Suppose $g \circ f(x) = (g \circ f)(y)$

We need to show $x = y$

Call $f(x)$ by the name X and $f(y)$ by the name Y

Since $g \circ f(x) = g \circ f(y)$ We have

$g(X) = g(f(x)) = g(f(x)) = g(Y)$

thus $X = Y$ since g is an injection

But that means $f(x) = f(y)$

So $x = y$ since f is an injection

Easy

1. $|A| \leq |B|$ and $|B| \leq |C|$ implies $|A| \leq |C|$
2. $|A| = |B|$ and $|B| = |C|$ implies $|A| = |C|$
3. $|A| \leq |B|$ and $|B| = |C|$, then $|A| \leq |C|$

etc.

Hard (Called: Shroder-Bernstein)

If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$

Example

$|\mathbb{N}| \leq |\mathbb{Z}|$

since $f : \mathbb{N} \to \mathbb{Z}$

given by $f(x) = x$ is an injection

Also $|Z| \leq |\mathbb{N}|$

Since $g : \mathbb{Z} \to \mathbb{N}$

given by $g(x) = 4x^2 + (-1)^x$ is an injection

There is no obvious way to combine f and g to form a bijection, but the theorem says it is possible.

Definition

If $|A| = |\mathbb{N}|$ we say
A is countably infinite

We may write

$$|A| = \aleph$$

(aleph-naught or the smallest possible infinity)

Note

A set is countably infinite iff, its elements can be listed as a sequence.

If $f : A \to \mathbb{N}$ is a bijection, we can list $f^{-1}(o), f^{-1}(1), f^{-1}(2)...$

Conversely, if $a_0, a_1, a_2, ...$

is a omplete list of elements of A, then $f : \mathbb{N} \to A$

given by $f(n) = a_n$

is a bijection.

In fact

A list permiting duplicates is fine, since we can erase the duplicates before applying numbers.

Briefly, we saw

$|\mathbb{N}| = |\mathbb{Z}^+|$

$|\mathbb{N}| = |\mathbb{Z}|$

$|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$

Our listing of $\mathbb{N}$ gurantess we can list $\mathbb{Q}$ (There is a whole thing he drew on the board, i cant be bothered trying to type it down)

Since

$|\mathbb{Z}| = |\mathbb{N}| = |\mathbb{Q}| = |\mathbb{N} \times \mathbb{N}|$ etc

Q: Is it possible all infinite sets are the same size?
A: No!

Cantor showed that no matter what set S we start with,
$|S| \leq |\mathcal{P}(S)|$

but

$|S| \neq |\mathcal{P}(S)|$

this is called a **diagonilization arguement**

Cool fact

$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$

But it is impossible for $f : \mathbb{N} \to \mathbb{R}$ to be a bijection.

Why its impossible

f cannot be a surjection

i.e there is at least one element of $\mathbb{R}$ not in the range of f

Given any function, $f : \mathbb{N} \to \mathbb{R}$

We can build a number $a_f \in [0, 1]$ not in the range of f.

Write a_f as $0.d_0, d_2, d_3 ... d_n$

i.e d_i is the first digit after the decimal

Pick d_i by looking at $f(i)$ and getting the $i + 1$st digit as wrong as possible.

If the $i + 1$st digit of $f(i)$ is x then:

$d_i = x + 5$ if $x \leq 4$

$d_i = x - 5$ if $x \geq 5$

Example

Suppose f lists:

$f(0) = 452.7653480....$

$f(1) = -78.2379641....$

$f(2) = 0.590148....$

$f(3) = 231.0004375....$

Our number a_f is $0.2858....$

Our number is not $f(i)$ for any i because its $i + 1$st decimal is wrong.